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I Multiple choice questions

1. The dimension of $\frac{1}{\mu_o \varepsilon_o}$ is

(a) [L T^{-1}] (b) [L^2 T^{-2}]

(c) [L^{-1} T] (d) [L^{-2} T^2]

2. If the amplitude of the magnetic field is 3×10^{-6} T, then amplitude of the electric field for a electromagnetic waves is

(a) 100 V m^{-1} (b) 300 V m^{-1}

(c) 600 V m^{-1} (d) 900 V m^{-1}

3. Which of the following electromagnetic radiation is used for viewing objects through fog

(a) microwave (b) gamma rays

(c) X- rays (d) infrared

4. Which of the following are false for electromagnetic waves

(a) transverse

(b) mechanical waves

(c) longitudinal

(d) produced by accelerating charges

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5. Consider an oscillator which has a charged particle and oscillates about its mean position with a frequency of 300 MHz. The wavelength of electromagnetic waves produced by this oscillator is

(a) 1 m (b) 10 m

(c) 100 m (d) 1000 m

6. The electric and the magnetic field, associated with an electromagnetic wave, propagating along X axis can be represented by

(a) $\vec{E} = E_o \hat{j}$ and $\vec{B} = B_o \hat{k}$

(b) $\vec{E} = E_o \hat{k}$ and $\vec{B} = B_o \hat{j}$

(c) $\vec{E} = E_o \hat{i}$ and $\vec{B} = B_o \hat{j}$

(d) $\vec{E} = E_o \hat{j}$ and $\vec{B} = B_o \hat{i}$

7. In an electromagnetic wave in free space the rms value of the electric field is 3 V m^{-1}. The peak value of the magnetic field is

(a) 1.414×10^{-8} T (b) 1.0×10^{-8} T

(c) 2.828×10^{-8} T (d) 2.0×10^{-8} T

8. During the propagation of electromagnetic waves in a medium:

(a) electric energy density is double of the magnetic energy density

(b) electric energy density is half of the magnetic energy density

(c) electric energy density is equal to the magnetic energy density

(d) both electric and magnetic energy densities are zero

9. If the magnetic monopole exists, then which of the Maxwell's equation to be modified?.

(a) $\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enclosed}}{\in_o}$

(b) $\oint \vec{E} \cdot d\vec{A} = 0$

(c) $\oint \vec{E} \cdot d\vec{A} = \mu_o I_{enclosed} + \mu_o \in_o \frac{d}{dt} \int \vec{E} \cdot d\vec{A}$

(d) $\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \Phi_B$

10. A radiation of energy E falls normally on a perfectly reflecting surface. The momentum transferred to the surface is

(a) $\frac{E}{c}$ (b) $2\frac{E}{c}$

(c) Ec (d) $\frac{E}{c^2}$

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